Hands- On Exam 1: Billing Module for POS (contributed by John Santillana)

The BnB Breakfast Cafe is a popular cafe to many people because of the high quality breakfast that they serve. They offer a variety of breakfast foods which people can't get enough of. Some of their menu items would also be on sale.

Here is a sample menu of their items:

ITEM CODE	ITEM	PRICE	DISCOUNT
BT	Beef Tapa w/ rice	80.00	0
BE	Bacon & Eggs w/ rice	95.50	0
PS	Pancake & Sausages	75.50	10%
CW	Chicken & Waffles	85.50	20%
OM	Omelette Special	60.00	0
GS	German Sausage & rice	99.75	20%
CB	Crispy Bangus w/ rice	65.00	10%
BB	Big Breakfast Meal	120.00	25%

So as not to keep customers waiting in line too long, a customer can only have a maximum order of four (4) kinds of menu items.

Here is a sample order:

ITEM CODE	QTY
GS	3
BB	1
BE	1
OM	2

You are tasked with creating a billing module for their Point of Sale(POS) system. It would take the customer's order and then compute and print out the customer's bill.

Task 1 (10 pts): Implement the displayItems() function that lists down the contents of the menu. ie. Item Code, Item Name, Price, Discount. Follow the format shown in the sample run below.

Task 2 (5 pts): Implement the initCustOrders() function that initializes the elements of sCustOrders[] to an empty string.

Task 3 (10 pts): Implement the getItemIndex() function. This will be needed to check if the customer's order (Item Code) exists in the sMenuItems otherwise it will return a -1.

Task 4 (12.5 pts): Implement the takeOrders() function. This function will ask the customer to input their order details, i.e. item code and quantity of order. The customer can order a maximum of 4 kinds of menu items(Repeat orders are allowed, meaning you can use the same item code more than once). The ordering is done either after the customer enters his/her 4th item or if the customer enters the letters "ED". Only valid orders will be stored in sCustOrders[] and fCustOrderDetail[]. Any invalid entry, i.e. non-existent item codes, invalid quantity will be ignored.

Task 5 (12.5 pts): implement the printBill() function. This function will print the list of orders. For each item, display the quantity, unit price, discount and sub-total. The sub-totals of each order will be stored in the fSubTotal array. The total amount due will be displayed as well. Follow the format shown in the sample run below. If there are zero(0) valid orders, a message will be displayed stating that there is no bill.

Assumptions:

- All item code inputs will be in uppercase letters.
- There will always only be a total of 4 orders regardless of whether the item code is invalid or not.
- Quantity will always be greater than 0.

FILES PROVIDED:

- **BILLING-MODULE-LASTNAME.c** – skeleton file which contains some initial code that you'll need to complete.
- **BILLING-MODULE-Main.c** – contains the main() functions for testing purposes
- **INPUT1.txt** - example input data file that contains the inputs seen in sample run 1
- **EXPECTED1.txt** – contains the expected result seen in sample run 1
- **INPUT2.txt** - example input data file that contains the inputs seen in sample run 2
- **EXPECTED2.txt** – contains the expected result seen in sample run 2
- **INPUT3.txt** - example input data file that contains the inputs seen in sample run 3
- **EXPECTED3.txt** – contains the expected result seen in sample run 3

Sample Run 1: All valid item codes

```
Welcome to BnB Breakfast Cafe!

ITEM CODE       |       ITEM NAME                       |       ITEM PRICE      |       DISCOUNT
 [BT]           |        Beef Tapa w/ rice              |        80.00          |        0.00%
 [BE]           |        Bacon & Eggs w/ rice           |        95.50          |        0.00%
 [PS]           |        Pancake & Sausages             |        75.50          |        10.00%
 [CW]           |        Chicken & Waffles              |        85.50          |        20.00%
 [OM]           |        Omelette Special               |        60.00          |        0.00%
 [GS]           |        German Sausage & rice          |        99.75          |        20.00%
 [CB]           |        Crispy Bangus w/ rice          |        65.00          |        10.00%
 [BB]           |        Big Breakfast Meal             |        120.00         |        25.00%

Please tell me your order!

Item Code: GS
Quantity: 3


Item Code: BB
Quantity: 1


Item Code: BE
Quantity: 1


Item Code: OM
Quantity: 2

====
BILL
====
        ITEM            |       QTY     |       PRICE           |       DISCOUNT        |       SUB-TOTAL
German Sausage & rice   |        3      |       99.75           |       19.95           |       239.40
Big Breakfast Meal      |        1      |       120.00          |       30.00           |       90.00
Bacon & Eggs w/ rice    |        1      |       95.50           |       0.00            |       95.50
Omelette Special        |        2      |       60.00           |       0.00            |       120.00

AMT. DUE:       544.90
```

Sample Run 2: 1 valid item code, 1 invalid item code, ED

```
Welcome to BnB Breakfast Cafe!

ITEM CODE        |        ITEM NAME                        |        ITEM PRICE       |        DISCOUNT
 [BT]            |         Beef Tapa w/ rice               |         80.00           |         0.00%
 [BE]            |         Bacon & Eggs w/ rice            |         95.50           |         0.00%
 [PS]            |         Pancake & Sausages              |         75.50           |         10.00%
 [CW]            |         Chicken & Waffles               |         85.50           |         20.00%
 [OM]            |         Omelette Special                |         60.00           |         0.00%
 [GS]            |         German Sausage & rice           |         99.75           |         20.00%
 [CB]            |         Crispy Bangus w/ rice           |         65.00           |         10.00%
 [BB]            |         Big Breakfast Meal              |         120.00          |         25.00%

Please tell me your order!

Item Code: PS
Quantity: 4


Item Code: XX
Quantity: 2


Item Code: ED

====
BILL
====
        ITEM            |        QTY      |        PRICE            |        DISCOUNT         |        SUB-TOTAL
Pancake & Sausages      |         4       |        75.50            |        7.55             |        271.80

AMT. DUE:       271.80
```

Sample Run 3: 2 valid item codes and 2 invalid item codes

```
Welcome to BnB Breakfast Cafe!

ITEM CODE        |        ITEM NAME                        |        ITEM PRICE       |        DISCOUNT
 [BT]            |         Beef Tapa w/ rice               |         80.00           |         0.00%
 [BE]            |         Bacon & Eggs w/ rice            |         95.50           |         0.00%
 [PS]            |         Pancake & Sausages              |         75.50           |         10.00%
 [CW]            |         Chicken & Waffles               |         85.50           |         20.00%
 [OM]            |         Omelette Special                |         60.00           |         0.00%
 [GS]            |         German Sausage & rice           |         99.75           |         20.00%
 [CB]            |         Crispy Bangus w/ rice           |         65.00           |         10.00%
 [BB]            |         Big Breakfast Meal              |         120.00          |         25.00%

Please tell me your order!

Item Code: BE
Quantity: 3


Item Code: XX
Quantity: 2


Item Code: GS
Quantity: 2


Item Code: YY
Quantity: 4

====
BILL
====
        ITEM            |        QTY      |        PRICE            |        DISCOUNT         |        SUB-TOTAL
Bacon & Eggs w/ rice    |         3       |        95.50            |        0.00             |        286.50
German Sausage & rice   |         2       |        99.75            |        19.95            |        159.60

AMT. DUE:       446.10
```

Sample Run 4: Repeating Item Code, Valid Item Code, End

```
Welcome to BnB Breakfast Cafe!

ITEM CODE       |       ITEM NAME                       |       ITEM PRICE      |       DISCOUNT
 [BT]           |        Beef Tapa w/ rice              |        80.00          |        0.00%
 [BE]           |        Bacon & Eggs w/ rice           |        95.50          |        0.00%
 [PS]           |        Pancake & Sausages             |        75.50          |        10.00%
 [CW]           |        Chicken & Waffles              |        85.50          |        20.00%
 [OM]           |        Omelette Special               |        60.00          |        0.00%
 [GS]           |        German Sausage & rice          |        99.75          |        20.00%
 [CB]           |        Crispy Bangus w/ rice          |        65.00          |        10.00%
 [BB]           |        Big Breakfast Meal             |        120.00         |        25.00%

Please tell me your order!

Item Code: BT
Quantity: 3


Item Code: BT
Quantity: 2


Item Code: OM
Quantity: 1


Item Code: ED

====
BILL
====
        ITEM            |       QTY     |       PRICE           |       DISCOUNT        |       SUB-TOTAL
Beef Tapa w/ rice       |        3      |       80.00           |       0.00            |       240.00
Beef Tapa w/ rice       |        2      |       80.00           |       0.00            |       160.00
Omelette Special        |        1      |       60.00           |       0.00            |       60.00

AMT. DUE:       460.00
```

Sample Run 4: All Invalid Item Codes

```
Welcome to BnB Breakfast Cafe!

ITEM CODE       |       ITEM NAME                       |       ITEM PRICE      |       DISCOUNT
 [BT]           |        Beef Tapa w/ rice              |        80.00          |        0.00%
 [BE]           |        Bacon & Eggs w/ rice           |        95.50          |        0.00%
 [PS]           |        Pancake & Sausages             |        75.50          |        10.00%
 [CW]           |        Chicken & Waffles              |        85.50          |        20.00%
 [OM]           |        Omelette Special               |        60.00          |        0.00%
 [GS]           |        German Sausage & rice          |        99.75          |        20.00%
 [CB]           |        Crispy Bangus w/ rice          |        65.00          |        10.00%
 [BB]           |        Big Breakfast Meal             |        120.00         |        25.00%

Please tell me your order!

Item Code: WW
Quantity: 1


Item Code: XX
Quantity: 2


Item Code: YY
Quantity: 3


Item Code: ZZ
Quantity: 4

NO BILL SINCE THERE ARE NO VALID ORDERS MADE!
```

Sample Run 5: Input ED at the start

```
Welcome to BnB Breakfast Cafe!

ITEM CODE       |       ITEM NAME                       |       ITEM PRICE      |       DISCOUNT
 [BT]           |        Beef Tapa w/ rice              |        80.00          |        0.00%
 [BE]           |        Bacon & Eggs w/ rice           |        95.50          |        0.00%
 [PS]           |        Pancake & Sausages             |        75.50          |        10.00%
 [CW]           |        Chicken & Waffles              |        85.50          |        20.00%
 [OM]           |        Omelette Special               |        60.00          |        0.00%
 [GS]           |        German Sausage & rice          |        99.75          |        20.00%
 [CB]           |        Crispy Bangus w/ rice          |        65.00          |        10.00%
 [BB]           |        Big Breakfast Meal             |        120.00         |        25.00%

Please tell me your order!

Item Code: ED

NO BILL SINCE THERE ARE NO VALID ORDERS MADE!
```

DELIVERABLES: Submit/upload two files before the Canvas submission deadline

1. **BILLING-MODULE-LASTNAME.c** -- this file contains your solution (i.e., C source code)
2. **`ACTUAL-LASTNAME.txt`** -- this is the actual output redirected text file produced by your own exe file. You can produce this file by running your a.exe file with INPUT and OUTPUT redirection in the command line interface as follows:

   prompt> **`a < INPUT1.txt > ACTUAL-LASTNAME.txt`** **(Press Enter)**

Make sure to rename these files using your own last name. For example, if your last name is SANTOS, you must upload your files as RESERVE-SANTOS.c and ACTUAL-SANTOS.c.

TESTING & SCORING:

- Your program will be tested using a different input text file (containing different words).
- Only codes, functions and syntax that have been discussed so far in the CCPROG series along with the course notes provided can be used in the exam.
- Points allocated for each correct function definition is indicated in the previous pages. Thus, the maximum total score will be 50/50.
- There will be a -1 deduction for formatting if the spacing and tabs are not followed.
- **A function that has a syntax/compilation error will result in a score of 0.**
- **A function that does not produce any of the required output (for example, due to runtime error) will result in a score of 0.**

----------NOTHING FOLLOWS----------